

Lecture 8 : Differentiation

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1. Differentiation

Let f be a function defined for all real value of x in an interval I . For $a \in I$, if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ or } \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

exists, then this limit is called the **differential coefficient of $f(x)$** at the point $x = a$. It is generally denoted by $f'(a)$.

The differential coefficient is also called the **derivative** or **derived function**. The process of finding the differential coefficient is called **differentiation**. In simple terms, the derivative of a function measures the rate of change of the function.

2. Notation

The differential coefficient of $f(x)$ is written generally as $\frac{d}{dx}f(x)$ or $f'(x)$ or $Df(x)$. the differential coefficient at the point $x = a$ is generally represented by $f'(a)$ or $\left\{ \frac{df(x)}{dx} \right\}_{x=a}$.

3. Left hand and right hand derivative

If

$$\lim_{h \rightarrow 0^-} \frac{f(a-h) - f(a)}{-h} \text{ or } \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a},$$

exists, then it is called the **left hand derivative** of f at a and is denoted by $f'(a-0)$ or $Lf'(a)$.

Again, if

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \text{ or } \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a},$$

exists, then it is called the **right hand derivative** of f at a and is denoted by $f'(a+0)$ or $Rf'(a)$.

Clearly, a function f is differentiable at $x = a$, if $Lf'(a)$, $Rf'(a)$ both exist and are equal.

4. Continuity and differentiation

The relation between continuity and differentiation is given by the theorem below :

Theorem 1 *If a function is differentiable at a point, then it is continuous at that point but the converse is not necessarily true.*

Proof: Let $f(x)$ be a differentiable at $x = a$, then $f'(a)$ exists and

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

We can write $f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a)$, $x \neq a$.

Taking limits on both sides as $x \rightarrow a$, we get

$$\lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) = f'(a) \cdot 0 = 0,$$

i.e., $\lim_{x \rightarrow a} f(x) = f(a)$, which proves that f is necessarily continuous at $x = a$, if it is differentiable at a .

Conversely, Consider the modulus function at the point $x = 0$, defined as follows :

$$f(x) = |x| = \begin{cases} x & \text{when } x > 0 \\ 0 & \text{when } x = 0 \\ -x & \text{when } x < 0. \end{cases}$$

Since, limit of f as $x \rightarrow 0$ from left and right is 0 and $f(0) = 0$. Hence, function f , is continuous at $x = 0$.

But at $x = 0$

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

and

$$Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0 - h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{|-h| - |0|}{-h} = \lim_{h \rightarrow 0} \frac{h}{(-h)} = -1.$$

Since, $Rf'(0) \neq Lf'(0)$, thus $f(x)$ is not differentiable at $x = 0$.

Example : Test the continuity and differentiability of the function

$$f(x) = \begin{cases} 1 + x & \text{when } x < 2 \\ 5 - x & \text{when } x \geq 2 \end{cases}$$

at $x = 2$.

Solution : At $x = 2$, the value of the function is $f(2) = 5 - 2 = 3$.

Now, $\lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} [5 - (2+h)] = 3$.

$\lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} [1 + (2-h)] = 3$.

Since, $\lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} f(2-h) = 3$, thus f is continuous at $x = 2$.

For differentiability of f at $x = 2$, we have

$$\begin{aligned} Rf'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[5 - (2+h)] - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} = -1. \end{aligned}$$

Again,

$$\begin{aligned} Lf'(2) &= \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{[1 + (2-h)] - 3}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{-h} = 1. \end{aligned}$$

Since, $Rf'(2) \neq Lf'(2)$, thus f is not differentiable at $x = 2$.

5. Chain rule of differentiation

Suppose that, we have two functions $f(x)$ and $g(x)$ and they are both differentiable., then

1. If we define $F(x) = (f \circ g)(x)$, then the derivative of $F(x)$ is,

$$F'(x) = f'(g(x))g'(x).$$

2. If we have $y = f(u)$ and $u = g(x)$, then the derivative of y is,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

Examples

1. Find the differentiation of $(ax^2 + b)^3$.

Solution: Put $(ax^2 + b)^3 = t$. Then,

$$\begin{aligned} \frac{d}{dx}(ax^2 + b)^3 &= \frac{d}{dx}(t^3) = \frac{d}{dt}(t^3) \frac{dt}{dx} = 3t^2 \frac{d}{dx}(ax^2 + b) \\ &= 3(ax^2 + b)^2 \cdot (2ax) = 6ax(ax^2 + b)^2. \end{aligned}$$

2. Find the differentiation of $\sin x^2$.

Solution: Put $x^2 = t$. Then,

$$\begin{aligned}\frac{d}{dx} \sin(x^2) &= \frac{d}{dx}(\sin t) = \frac{d}{dt}(\sin t) \frac{dt}{dx} = \cos t \frac{d}{dx}(x^2) \\ &= 2x \cos x^2.\end{aligned}$$

Reference book : *Introduction to real analysis by S.K. Mapa.*